Refraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a change in wave speed to wavelength and direction.

SUCCESS CRITERIA - I can...

1. Measure ray angles from the normal.
2. Explain frequency continuity at a stationary boundary.
3. Predict bending toward or away from the normal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “refraction” and “normal line” to explain your model or result. Sentence starters are optional.

refraction: change in wave direction when speed changes at a boundary

normal line: a reference line perpendicular to a surface

medium: the material through which a mechanical wave travels

01 Warm-up: bring back an earlier idea

A wave snapshot has crests at 0.1, 0.4, 0.7 m.
Find wavelength. Its period is 0.05 s; find frequency.

Word help: wavelength = distance between corresponding points on repeated waves

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

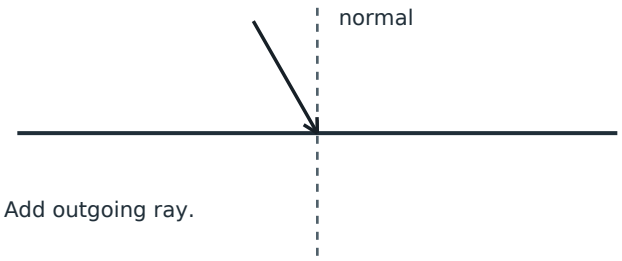
02 Worked example: explain the model

Reference: $n_1 \sin \theta_1 = n_2 \sin \theta_2$; $n = c/v$; $v = f \lambda$.
Angles from the normal. Stationary boundary: frequency stays constant.

Worked reasoning: Light enters a slower medium at an angle to the normal and bends toward the normal. The normal is perpendicular to the boundary.

Your turn: Draw and label the boundary, normal, incident ray and refracted ray.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Marzano check - circle: ① ② ③

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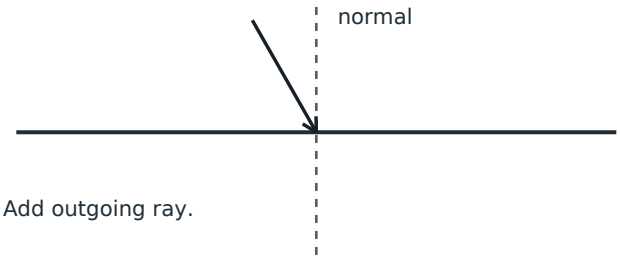
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Mark the normal at the boundary. A ray enters a slower medium at an angle.
Sketch the refracted direction.

Sentence starter: The ... in my model represents ... . Its label shows ... .

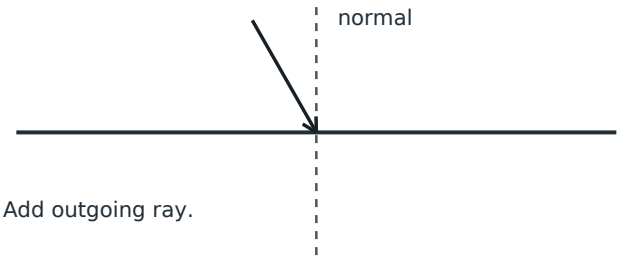


Marzano check - circle: 1 2 3

04 Apply the physics

A laser ray travels from air (n = 1.00) into glass (n = 1.50) at 30° to the normal.
Label both media and angles, then use $n_1 \sin \theta_1 = n_2 \sin \theta_2$ to find the refracted angle.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

A student measures an incidence angle from the surface instead of the normal. If surface angle is 20°, what angle belongs in Snell's law?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 0.4 kg metal cools from 92 to 32°C while 0.3 kg water warms from 12 to 32°C.
Use water c=4200; neglect cup/losses.
Find metal c.
Label temperatures, samples and relevant energy transfers.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

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Continue down each column.

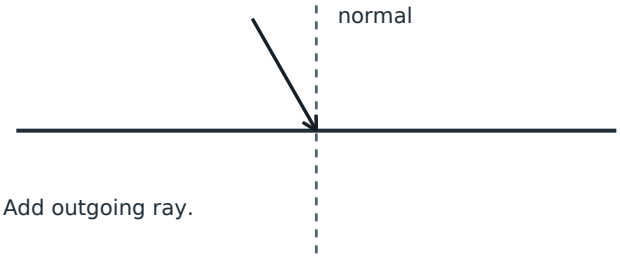
Sentence starters are optional. You may plan in any language.

07 Exit check

From glass (n=1.5) to air, find critical angle using sin θ_c=1/1.5.
State what happens above that angle in ideal model.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



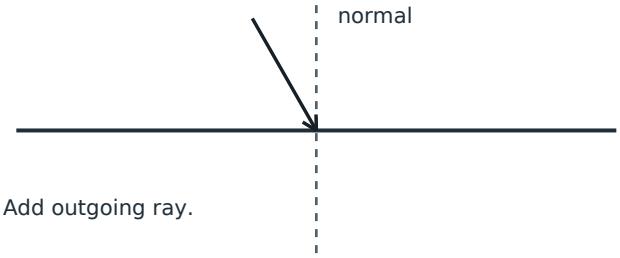
Add outgoing ray.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A ray travels from water (n = 1.33) into glass (n = 1.50) at 40° to the normal.
Use $n_1 \sin \theta_1 = n_2 \sin \theta_2$ to find the refracted angle. Then find the critical angle for travel from this glass into water.
Label both cases.

Sentence starter: I found … [unit]. This value means … in this situation.



Add outgoing ray.

Marzano check - circle: 1 2 3